

Similarity and Dissimilarity of Data Objects

Similarity and Dissimilarity of Data Objects

- similarity and dissimilarity tell us how close or how different two data objects are (two records, two customers, two documents, etc.). These measures are used in clustering, classification, recommendation, anomaly detection, etc.

1) Similarity

- Similarity means how much two objects are alike.
- Higher similarity $\Rightarrow$ more alike
- Usually ranges from 0 to 1
 - 1 = exactly same
 - 0 = totally different

2) Dissimilarity

- Dissimilarity means how much two objects are different.
- Higher dissimilarity $\Rightarrow$ more different
- Often represented using distance measures
 - 0 = exactly same
 - Larger value = more different

- Similarity measure gives how close / alike two objects are
Higher value = more similar
- Dissimilarity (distance) measure gives how far / different two objects are
Higher value = more dissimilar

1) Euclidean Distance (Dissimilarity)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Output:

- 0 means exactly same
- bigger value means more different

Calculate the distance between the points (4,1) and (3,0).

- *Using Euclidean Distance Formula:*
- $\Rightarrow d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
 $\Rightarrow d = \sqrt{(3 - 4)^2 + (0 - 1)^2}$
 $\Rightarrow d = \sqrt{1 + 1}$
 $\Rightarrow d = \sqrt{2} = 1.414 \text{ unit}$

2) Manhattan Distance (Dissimilarity)

Formula

For two points $P = (x_1, y_1)$, $Q = (x_2, y_2)$, the Manhattan distance is:

- For 2-dimensions: $|x_1 - x_2| + |y_1 - y_2|$

- 0 → exactly same
- bigger value → more different

Points $P = (1, 2), Q = (4, 0)$

$$D(P, Q) = |x_1 - x_2| + |y_1 - y_2| = |1 - 4| + |2 - 0| = 3 + 2 = 5$$

Cosine Similarity (Similarity)

- *Cosine Similarity* = $\frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum A_i B_i}{\sqrt{\sum A_i^2} \sqrt{\sum B_i^2}}$

- $X1=(1.5,1.7), X2=(1.4,1.6)$

$$\frac{(1.4 * 1.5) + (1.6 * 1.7)}{\sqrt{(1.5^2 + 1.7^2)} * \sqrt{(1.4^2 + 1.6^2)}} = 0.9999$$

Supremam Distance

- Let $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$
- Supremum Distance = $\max(|x_2 - x_1|, |y_2 - y_1|)$

- $X1=(1.5,1.7), X2=(1.4,1.6)$
- Supremam Distance $=\max(|1.4-1.5|, |1.6-1.7|)=0.1$

Example

- Suppose we have the following two-dimensional data set:
- Consider the data as two-dimensional data points.
- Given a new data point, $x = (1.4, 1.6)$ as a query, rank the database points based on similarity with the query using Euclidean distance, Manhattan distance, ~~Minkowski distance~~ supremum distance, and cosine similarity.

	A_1	A_2
x_1	1.5	1.7
x_2	2	1.9
x_3	1.6	1.8
x_4	1.2	1.5
x_5	1.5	1.0

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	A_1	A_2	Euclidian Distance
1	1.5	1.7	$\sqrt{(1.4 - 1.5)^2 + (1.6 - 1.7)^2} = 0.1414$
2	2	1.9	$\sqrt{(1.4 - 2.0)^2 + (1.6 - 1.9)^2} = 0.6708$
3	1.6	1.8	$\sqrt{(1.4 - 1.6)^2 + (1.6 - 1.6)^2} = 0.2828$
4	1.2	1.5	$\sqrt{(1.4 - 1.2)^2 + (1.6 - 1.5)^2} = 0.2236$
5	1.5	1.0	$\sqrt{(1.4 - 1.5)^2 + (1.6 - 1.0)^2} = 0.6083$

$x = (1.4, 1.6)$

- Let $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$
- Euclidian Distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

	A_1	A_2	Manhattan Distance
1	1.5	1.7	$ 1.4 - 1.5 + 1.6 - 1.7 = 0.2$
2	2	1.9	$ 1.4 - 2.0 + 1.6 - 1.9 = 0.9$
3	1.6	1.8	$ 1.4 - 1.6 + 1.6 - 1.8 = 0.4$
4	1.2	1.5	$ 1.4 - 1.2 + 1.6 - 1.5 = 0.3$
5	1.5	1.0	$ 1.4 - 1.5 + 1.6 - 1.0 = 0.7$

$x = (1.4, 1.6)$

- Let $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$
- $\text{Manhattan Distance} = |x_2 - x_1| + |y_2 - y_1|$

	A_1	A_2	Supremum Distance
1	1.5	1.7	$\max((1.4 - 1.5) , (1.6, -1.7)) = 0.1$
2	2	1.9	$\max((\underline{1.4 - 2.0}) , (\underline{1.6, -1.9})) = 0.6$
3	1.6	1.8	$\max((1.4 - 1.6) , (1.6, -1.8)) = 0.2$
4	1.2	1.5	$\max((1.4 - 1.2) , (1.6, -1.5)) = 0.2$
5	1.5	1.0	$\max((1.4 - 1.5) , (1.6, -1.0)) = 0.6$

$x = (1.4, 1.6)$

- Let $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$
- *Supremum Distance* = $\max(|(x_2 - x_1)|, |(y_2 - y_1)|)$

	A_1	A_2	Cosine Similarity
1	1.5	1.7	0.99999
2	2	1.9	0.99575
3	1.6	1.8	0.99997
4	1.2	1.5	0.99903
5	1.5	1.0	0.96536

$x = (1.4, 1.6)$

- $$\text{Cosine Similarity} = \frac{A \cdot B}{||A|| ||B||} = \frac{\sum A_i B_i}{\sqrt{\sum A_i^2} \sqrt{\sum B_i^2}}$$